In [26]:
```
import numpy as np
import pandas as pd
df=pd.read_csv("Book1.csv")
```

In [27]:
```
df.duplicated()
```

Out[27]:
```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9      True
10    False
dtype: bool
```

In [28]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 9 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   CustomerID       11 non-null     int64
 1   Age_Group        11 non-null     str
 2   Rating(1-5)      11 non-null     int64
 3   Hotel            11 non-null     str
 4   FoodPreference   11 non-null     str
 5   Bill             11 non-null     int64
 6   NoOfPax          11 non-null     int64
 7   EstimatedSalary  10 non-null     float64
 8   Age_Group.1      11 non-null     str
dtypes: float64(1), int64(4), str(4)
memory usage: 1.1 KB
```

In [29]:
```
df.drop_duplicates(inplace=True)
df
```

Out[29]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Estim
0	101	18-22	4	Taj	Veg	1450	2	
1	102	23-27	5	Marriott	Non-Veg	2100	3	
2	103	28-32	6	Hyatt	Veg	1580	2	
3	104	18-22	-1	Marriott	Vegetarian	1320	2	
4	105	33+	3	Taj	Veg	980	2	
5	106	33+	4	Tajj	Non-Veg	1950	2	
6	107	33+	4	Hyatt	Vegetarian	1100	-1	
7	108	18-22	7	Marriott	Veg	3100	-10	
8	109	28-32	2	Taj	Non-Veg	3600	3	
10	110	23-27	5	Hyatt	non-Veg	4	87700	

In [30]:
```
len(df)
```

Out[30]:
```
10
```

In [31]:
```
index=np.array(list(range(0,len(df))))
df.set_index(index,inplace=True)
index
```

Out[31]:
```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

In [32]:
```
df.drop(['Age_Group.1'],axis=1,inplace=True)
df
```

Out[32]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Estim
0	101	18-22	4	Taj	Veg	1450	2	
1	102	23-27	5	Marriott	Non-Veg	2100	3	
2	103	28-32	6	Hyatt	Veg	1580	2	
3	104	18-22	-1	Marriott	Vegetarian	1320	2	
4	105	33+	3	Taj	Veg	980	2	
5	106	33+	4	Tajj	Non-Veg	1950	2	
6	107	33+	4	Hyatt	Vegetarian	1100	-1	
7	108	18-22	7	Marriott	Veg	3100	-10	
8	109	28-32	2	Taj	Non-Veg	3600	3	
9	110	23-27	5	Hyatt	non-Veg	4	87700	

In [33]:
```
df.loc[df['Bill'] <0,'Bill']=np.nan
df
```

Out[33]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Esti
0	101	18-22	4	Taj	Veg	1450.0	2	
1	102	23-27	5	Marriott	Non-Veg	2100.0	3	
2	103	28-32	6	Hyatt	Veg	1580.0	2	
3	104	18-22	-1	Marriott	Vegetarian	1320.0	2	
4	105	33+	3	Taj	Veg	980.0	2	
5	106	33+	4	Tajj	Non-Veg	1950.0	2	
6	107	33+	4	Hyatt	Vegetarian	1100.0	-1	
7	108	18-22	7	Marriott	Veg	3100.0	-10	
8	109	28-32	2	Taj	Non-Veg	3600.0	3	
9	110	23-27	5	Hyatt	non-Veg	4.0	87700	

In [34]:
```
df.loc[df['EstimatedSalary'] <0,'EstimatedSalary'] =np.nan
df
```

Out[34]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Esti
0	101	18-22	4	Taj	Veg	1450.0	2	
1	102	23-27	5	Marriott	Non-Veg	2100.0	3	
2	103	28-32	6	Hyatt	Veg	1580.0	2	
3	104	18-22	-1	Marriott	Vegetarian	1320.0	2	
4	105	33+	3	Taj	Veg	980.0	2	
5	106	33+	4	Tajj	Non-Veg	1950.0	2	
6	107	33+	4	Hyatt	Vegetarian	1100.0	-1	
7	108	18-22	7	Marriott	Veg	3100.0	-10	
8	109	28-32	2	Taj	Non-Veg	3600.0	3	
9	110	23-27	5	Hyatt	non-Veg	4.0	87700	

In [35]:
```
df.loc[(df['NoOfPax'] <1)|(df['NoOfPax'] >20),'NoOfPax']=np.nan
df
```

Out[35]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Esti
0	101	18-22	4	Taj	Veg	1450.0	2.0	
1	102	23-27	5	Marriott	Non-Veg	2100.0	3.0	
2	103	28-32	6	Hyatt	Veg	1580.0	2.0	
3	104	18-22	-1	Marriott	Vegetarian	1320.0	2.0	
4	105	33+	3	Taj	Veg	980.0	2.0	
5	106	33+	4	Tajj	Non-Veg	1950.0	2.0	
6	107	33+	4	Hyatt	Vegetarian	1100.0	NaN	
7	108	18-22	7	Marriott	Veg	3100.0	NaN	
8	109	28-32	2	Taj	Non-Veg	3600.0	3.0	
9	110	23-27	5	Hyatt	non-Veg	4.0	NaN	

In [36]:
```
df.Age_Group.unique()
```

Out[36]:
```
<ArrowStringArray>
['18-22', '23-27', '28-32', '33+']
Length: 4, dtype: str
```

In [37]:
```
df.Hotel.unique()
```

Out[37]:
```
<ArrowStringArray>
['Taj', 'Marriott', 'Hyatt', 'Tajj']
Length: 4, dtype: str
```

In [42]:
```
df['Hotel']=df['Hotel'].replace('Tajj','Taj')
df
```

Out[42]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Esti
0	101	18-22	4	Taj	Veg	1450.0	2.0	
1	102	23-27	5	Marriott	Non-Veg	2100.0	3.0	
2	103	28-32	6	Hyatt	Veg	1580.0	2.0	
3	104	18-22	-1	Marriott	Vegetarian	1320.0	2.0	
4	105	33+	3	Taj	Veg	980.0	2.0	
5	106	33+	4	Taj	Non-Veg	1950.0	2.0	
6	107	33+	4	Hyatt	Vegetarian	1100.0	NaN	
7	108	18-22	7	Marriott	Veg	3100.0	NaN	
8	109	28-32	2	Taj	Non-Veg	3600.0	3.0	
9	110	23-27	5	Hyatt	non-Veg	4.0	NaN	

In [43]:
```
df.FoodPreference.unique()
```

Out[43]:
```
<ArrowStringArray>
['Veg', 'Non-Veg', 'Vegetarian', 'non-Veg']
Length: 4, dtype: str
```

In [45]:
```
df['FoodPreference']=df['FoodPreference'].replace(['Vegetarian','veg'] ,'Veg')
df
```

Out[45]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Esti
0	101	18-22	4	Taj	Veg	1450.0	2.0	
1	102	23-27	5	Marriott	Non-Veg	2100.0	3.0	
2	103	28-32	6	Hyatt	Veg	1580.0	2.0	
3	104	18-22	-1	Marriott	Veg	1320.0	2.0	
4	105	33+	3	Taj	Veg	980.0	2.0	
5	106	33+	4	Taj	Non-Veg	1950.0	2.0	
6	107	33+	4	Hyatt	Veg	1100.0	NaN	
7	108	18-22	7	Marriott	Veg	3100.0	NaN	
8	109	28-32	2	Taj	Non-Veg	3600.0	3.0	
9	110	23-27	5	Hyatt	non-Veg	4.0	NaN	

In [46]:
```
df['FoodPreference']=df['FoodPreference'].replace('non-Veg','Non-Veg')
df
```

Out[46]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Esti
0	101	18-22	4	Taj	Veg	1450.0	2.0	
1	102	23-27	5	Marriott	Non-Veg	2100.0	3.0	
2	103	28-32	6	Hyatt	Veg	1580.0	2.0	
3	104	18-22	-1	Marriott	Veg	1320.0	2.0	
4	105	33+	3	Taj	Veg	980.0	2.0	
5	106	33+	4	Taj	Non-Veg	1950.0	2.0	
6	107	33+	4	Hyatt	Veg	1100.0	NaN	
7	108	18-22	7	Marriott	Veg	3100.0	NaN	
8	109	28-32	2	Taj	Non-Veg	3600.0	3.0	
9	110	23-27	5	Hyatt	Non-Veg	4.0	NaN	

In [47]:
```
df['EstimatedSalary']=df['EstimatedSalary'].fillna(round(df['EstimatedSalary'].m
df
```

Out[47]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Esti
0	101	18-22	4	Taj	Veg	1450.0	2.0	
1	102	23-27	5	Marriott	Non-Veg	2100.0	3.0	
2	103	28-32	6	Hyatt	Veg	1580.0	2.0	
3	104	18-22	-1	Marriott	Veg	1320.0	2.0	
4	105	33+	3	Taj	Veg	980.0	2.0	
5	106	33+	4	Taj	Non-Veg	1950.0	2.0	
6	107	33+	4	Hyatt	Veg	1100.0	NaN	
7	108	18-22	7	Marriott	Veg	3100.0	NaN	
8	109	28-32	2	Taj	Non-Veg	3600.0	3.0	
9	110	23-27	5	Hyatt	Non-Veg	4.0	NaN	

In [48]:
```
df['NoOfPax'] =df['NoOfPax'].fillna(round(df['NoOfPax'].median()))
df
```

Out[48]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Esti
0	101	18-22	4	Taj	Veg	1450.0	2.0	
1	102	23-27	5	Marriott	Non-Veg	2100.0	3.0	
2	103	28-32	6	Hyatt	Veg	1580.0	2.0	
3	104	18-22	-1	Marriott	Veg	1320.0	2.0	
4	105	33+	3	Taj	Veg	980.0	2.0	
5	106	33+	4	Taj	Non-Veg	1950.0	2.0	
6	107	33+	4	Hyatt	Veg	1100.0	2.0	
7	108	18-22	7	Marriott	Veg	3100.0	2.0	
8	109	28-32	2	Taj	Non-Veg	3600.0	3.0	
9	110	23-27	5	Hyatt	Non-Veg	4.0	2.0	

In [49]:
```
df['Rating(1-5)']=df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()))
df
```

Out[49]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Esti
0	101	18-22	4	Taj	Veg	1450.0	2.0	
1	102	23-27	5	Marriott	Non-Veg	2100.0	3.0	
2	103	28-32	6	Hyatt	Veg	1580.0	2.0	
3	104	18-22	-1	Marriott	Veg	1320.0	2.0	
4	105	33+	3	Taj	Veg	980.0	2.0	
5	106	33+	4	Taj	Non-Veg	1950.0	2.0	
6	107	33+	4	Hyatt	Veg	1100.0	2.0	
7	108	18-22	7	Marriott	Veg	3100.0	2.0	
8	109	28-32	2	Taj	Non-Veg	3600.0	3.0	
9	110	23-27	5	Hyatt	Non-Veg	4.0	2.0	

In [50]:
```
df['Bill']=df['Bill'].fillna(round(df['Bill'].mean()))
df
```

Out[50]:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Esti
0	101	18-22	4	Taj	Veg	1450.0	2.0	
1	102	23-27	5	Marriott	Non-Veg	2100.0	3.0	
2	103	28-32	6	Hyatt	Veg	1580.0	2.0	
3	104	18-22	-1	Marriott	Veg	1320.0	2.0	
4	105	33+	3	Taj	Veg	980.0	2.0	
5	106	33+	4	Taj	Non-Veg	1950.0	2.0	
6	107	33+	4	Hyatt	Veg	1100.0	2.0	
7	108	18-22	7	Marriott	Veg	3100.0	2.0	
8	109	28-32	2	Taj	Non-Veg	3600.0	3.0	
9	110	23-27	5	Hyatt	Non-Veg	4.0	2.0	

In [ ]: